

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Obstacle Avoidance and Occlusion-Aware UAV Trajectory Optimization

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I. POSTER TITLE

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Obstacle Avoidance and Occlusion-Aware UAV Trajectory Optimization.

II. POSTER AUTHORS

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III. POSTER ABSTRACT

Trajectory optimization for UAV target tracking in cluttered environments must jointly handle obstacle avoidance, target visibility, and terminal-goal constraints. This poster presents a discrete Higher-Order Unconstrained Binary Optimization (HUBO) formulation for obstacle-avoidance and occlusion-aware UAV trajectory planning, together with a feasibility-aware evaluation protocol for soft-penalty quantum-optimization benchmarks. The work does not claim quantum speedup. Instead, it asks whether a higher-order binary formulation captures meaningful planning structure and how penalty-based solvers should be evaluated. Feasibility-aware benchmarking means that solver outputs are decoded as trajectories and checked for one-hot occupancy, start, motion, obstacle, and terminal constraints, rather than judged only by total penalized energy. On an 8×8 instance with 960 original binary variables, 87,684 monomials, and 3,172 cubic terms, we compare a classical A* planner, simulated annealing, D-Wave/Ocean neal after quadratization to a binary quadratic model, and Qiskit QAOA on a reduced quadratized instance. A* and simulated annealing obtain comparable best soft costs, but A* is far faster and always feasible. Neal is fast after quadratization but yields low decoded feasibility unless penalty weights are substantially increased. The poster highlights the need for constraint-aware validation when quantum-classical optimization methods are used for robotics-inspired planning benchmarks.

IV. POSTER RELEVANCE

The poster is relevant to QCE topics in hybrid quantum-classical computing, adiabatic quantum computation and quantum annealing, QAOA, variational and empirical algorithm analysis, quantum applications in optimization and decision making, and quantum systems software benchmarking. It connects a robotics-inspired planning problem to quantum-compatible optimization models while emphasizing practical validation: decoded feasibility, qubit/logical-variable growth after quadratization, solver-specific assumptions, and comparison with classical baselines.

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Abstract—We present a soft-penalty higher-order unconstrained binary optimization (HUBO) benchmark for UAV trajectory planning with obstacle avoidance and line-of-sight visibility. The goal is not to claim quantum speedup, but to define a quantum-optimization-compatible formulation and a feasibility-aware protocol for evaluating penalty-based solvers. Feasibility-aware means that decoded trajectories are checked for one-hot occupancy, start, motion, obstacle, and terminal constraints separately from the penalized objective value. The full 8×8 instance has 960 original binary variables and 3,172 cubic terms. A custom simulated annealing baseline evaluates the higher-order objective, while D-Wave/Ocean *neal* is used only after quadratization to a binary quadratic model (BQM). Qiskit QAOA is tested only on a reduced quadratized instance as a correctness/compatibility check. Results show comparable best soft costs for A* and simulated annealing, but only 22% true feasibility for *neal*/BQM at the hardened weight setting, motivating explicit feasibility checks.

Index Terms—HUBO, QUBO, quantum annealing, QAOA, A*, simulated annealing, UAV trajectory optimization, occlusion-aware tracking, feasibility-aware benchmarking.

I. INTRODUCTION AND PROBLEM SETTING

Visibility-aware UAV target tracking keeps a target in view while avoiding obstacles. Masnawi *et al.* formulated the corresponding occlusion-free target-tracking and obstacle-avoidance problem as a real-time multi-convex model predictive control (MPC) problem [1]. We use that work as the continuous-control reference and formulate a discrete soft-penalty benchmark for quantum-optimization workflows.

Let V be grid cells, $O \subset V$ obstacle cells, $B \subset V$ buffer cells near obstacles or visibility-critical regions, s the start cell, g_t the target cell at time t , and $N(u)$ the legal neighbor cells reachable from u in one step. A buffer cell is not necessarily an obstacle; it is a cell in which sustained residence is discouraged. The binary variable $x_{t,v} \in \{0, 1\}$ equals one iff the UAV occupies cell v at time t . The discrete planning problem we approximate is

$$\min_x C_{\text{vis}}(x) + C_{\text{prox}}(x) + C_{\text{term}}(x) \quad (1)$$

$$\text{s.t.} \quad \sum_v x_{t,v} = 1 \quad \forall t, \quad (2)$$

$$x_{0,s} = 1, \quad x_{t,v} = 0 \quad \forall v \in O, \quad (3)$$

$$x_{t,u}x_{t+1,v} = 0 \quad \forall v \notin N(u). \quad (4)$$

The continuous MPC dynamics and continuous visibility constraints are not reproduced because of space; the reader

should consult [1]. This benchmark replaces hard constraints by penalties and then checks feasibility independently after decoding.

II. HUBO REFORMULATION

The penalty Hamiltonian is

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}. \quad (5)$$

Here $H_{\text{uniq}} = \lambda_u \sum_t (\sum_v x_{t,v} - 1)^2$ enforces one UAV cell per time; $H_{\text{start}} = \lambda_s(1 - x_{0,s})$ fixes the start; $H_{\text{move}} = \lambda_m \sum_t \sum_{v \notin N(u)} x_{t,u}x_{t+1,v}$ penalizes illegal transitions; and $H_{\text{obs}} = \lambda_o \sum_t \sum_{v \in O} x_{t,v}$ forbids obstacle occupancy.

We define $\text{LOS}(u, g_t)$ as the set of grid cells intersected by the Bresenham line segment from UAV cell u to target cell g_t . Occlusion-aware means that obstacle intersections with this line of sight affect the cost:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_t \sum_{u \in V} x_{t,u} |\text{LOS}(u, g_t) \cap O|. \quad (6)$$

For a static target this term is linear in the UAV variables; for a moving target encoded with binary target occupancy, it becomes a joint product of UAV and target occupancy variables. The sustained-proximity term is genuinely cubic:

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{b \in B} \sum_{t=1}^{T-2} x_{t-1,b}x_{t,b}x_{t+1,b}. \quad (7)$$

Finally, H_{term} enforces or rewards terminal arrival at the target or a terminal region. The full 8×8 instance has 960 original variables and 3,172 cubic terms; a one-ancilla-per-cubic quadratization would yield about 4,132 logical BQM variables before hardware-embedding overhead.

III. FEASIBILITY-AWARE BENCHMARKING

This is a soft-penalty instance: a solver can obtain a low total penalized energy $H(x)$ while violating hard planning constraints. We use “raw energy” only to mean the total penalized objective before decomposition. Feasibility-aware benchmarking means that each sample is decoded into a trajectory and checked for one-hot occupancy, start, motion, obstacle, and terminal conditions. We report hard-penalty violations, best feasible soft cost, target-reaching, visibility, and runtime separately.

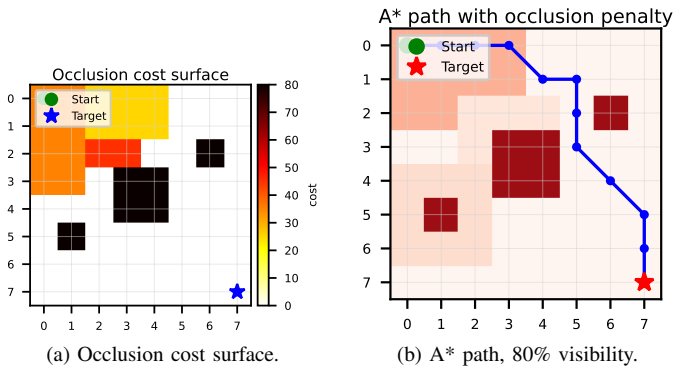


Fig. 1. Discrete occlusion-aware benchmark instance and a feasible A* trajectory.

TABLE I
FULL 8×8 INSTANCE, 960 ORIGINAL VARIABLES.

Metric	A*	SA	neal/BQM
Best soft cost	317	303	496
Feasibility rate	100%	100%	22%
Reaches target	yes	14%	8%
Visibility	80%	—	—
Runtime/sample	10 ms	47.8 s	0.26 s

The solvers are used in their appropriate regimes. A* is a classical graph-search baseline [4]. SA denotes a custom simulated annealing baseline applied to the higher-order benchmark objective [5]. D-Wave/Ocean neal is not a quantum solver and not a direct HUBO solver; it is a classical simulated-annealing BQM/QUBO sampler [7], so the higher-order model must be quadratized before it is sampled [8]. Qiskit QAOA [6], [9] is applied only to a quadratic representation on a tiny instance, not to the full HUBO. Direct k -local QAOA is possible in principle but was not the implementation tested here.

IV. RESULTS

Table I reports the matched comparison on the full 8×8 instance using 50 matched stochastic samples, SA at 500 sweeps, and hardened weights $\lambda = 1000$. A* and SA reach comparable best costs, but A* solves the planning analogue in about 10 ms while SA requires 47.8 s. Neal/BQM is fast after quadratization, but only 22% of its samples are feasible and its best feasible cost is worse.

On the reduced 2×2 demonstrator, the QAOA experiment uses eight logical binary variables and is included as a correctness/compatibility check. A*, SA, neal, and Qiskit QAOA all reach $E^* = -118.17$. QAOA takes 55 s versus 0.02 s for neal and 0.14 s for SA, so it is not presented as a scalable performance result.

V. DISCUSSION AND CONCLUSION

The A* baseline answers whether the binary formulation preserves sensible planning structure: it finds a feasible, target-reaching, visibility-aware trajectory in milliseconds. The value of the HUBO formulation is instead its compatibility with higher-order and quadratized quantum-optimization workflows and its extensibility to moving targets, where line-of-sight interactions can become intrinsically higher-order. The key

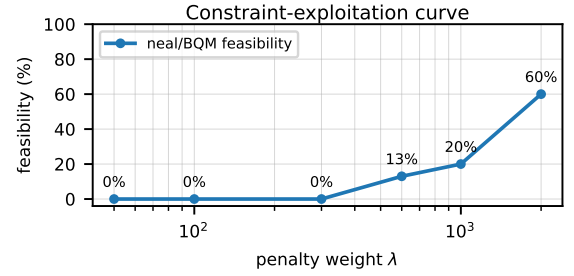


Fig. 2. Headline constraint-exploitation result: neal/BQM feasibility increases from 0% at $\lambda = 50$ to 60% at $\lambda = 2000$. Single-weight reports can therefore overstate solver quality.

methodological conclusion is that penalty-model benchmarks should be evaluated by decoded feasibility, not raw penalized energy alone.

Future work will verify and release the implementation, compare against the continuous MPC baseline of [1], test moving-target instances, and evaluate quadratized models on real quantum annealing hardware.

REFERENCES

- [1] H. Masnavi *et al.*, “Real-Time Multi-Convex MPC for Occlusion-Free Target Tracking With Quadrotors,” *IEEE Access*, vol. 10, pp. 113537–113550, 2022.
- [2] N. Innan *et al.*, “QUAV: Quantum-assisted path planning and optimization for UAV navigation with obstacle avoidance,” arXiv:2508.21361, 2025.
- [3] E. Osaba *et al.*, “Solving drone routing problems with quantum computing,” arXiv:2501.18432, 2025.
- [4] P. Hart, N. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Trans. Syst. Sci. Cybern.*, 1968.
- [5] S. Kirkpatrick, C. Gelatt, and M. Vecchi, “Optimization by simulated annealing,” *Science*, 1983.
- [6] E. Farhi, J. Goldstone, and S. Gutmann, “A quantum approximate optimization algorithm,” arXiv:1411.4028, 2014.
- [7] D-Wave Systems, “dwave-neal: SimulatedAnnealingSampler documentation,” Ocean SDK documentation.
- [8] D-Wave Systems, “dimod make_quadratic and BQM documentation,” Ocean SDK documentation.
- [9] Qiskit Community, “Qiskit: An open-source framework for quantum computing,” Zenodo, doi:10.5281/zenodo.256211, 2019.

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SCOPE

We introduce a quantum-compatible HUBO benchmark for occlusion-aware UAV trajectory optimization and a feasibility-aware methodology for evaluating such models. Classical baselines are deliberate: current quantum hardware limits make strong classical references essential, while the higher-order formulation and reduced-instance QAOA experiment chart a path toward future D-Wave / QAOA hardware studies.

1. The Problem

UAV target tracking in cluttered space must **simultaneously**:

- ▶ avoid obstacles, and
- ▶ keep **line-of-sight** to a target (occlusion-free).

Masnavi *et al.* solved this as continuous multi-convex MPC. We recast the **same problem class** as a discrete higher-order binary problem for quantum solvers.

2. HUBO Formulation

Binary $x[t, v] = 1$ iff UAV at cell v , time t . Seven cost terms:

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}$$

Occlusion (Bresenham line-of-sight):

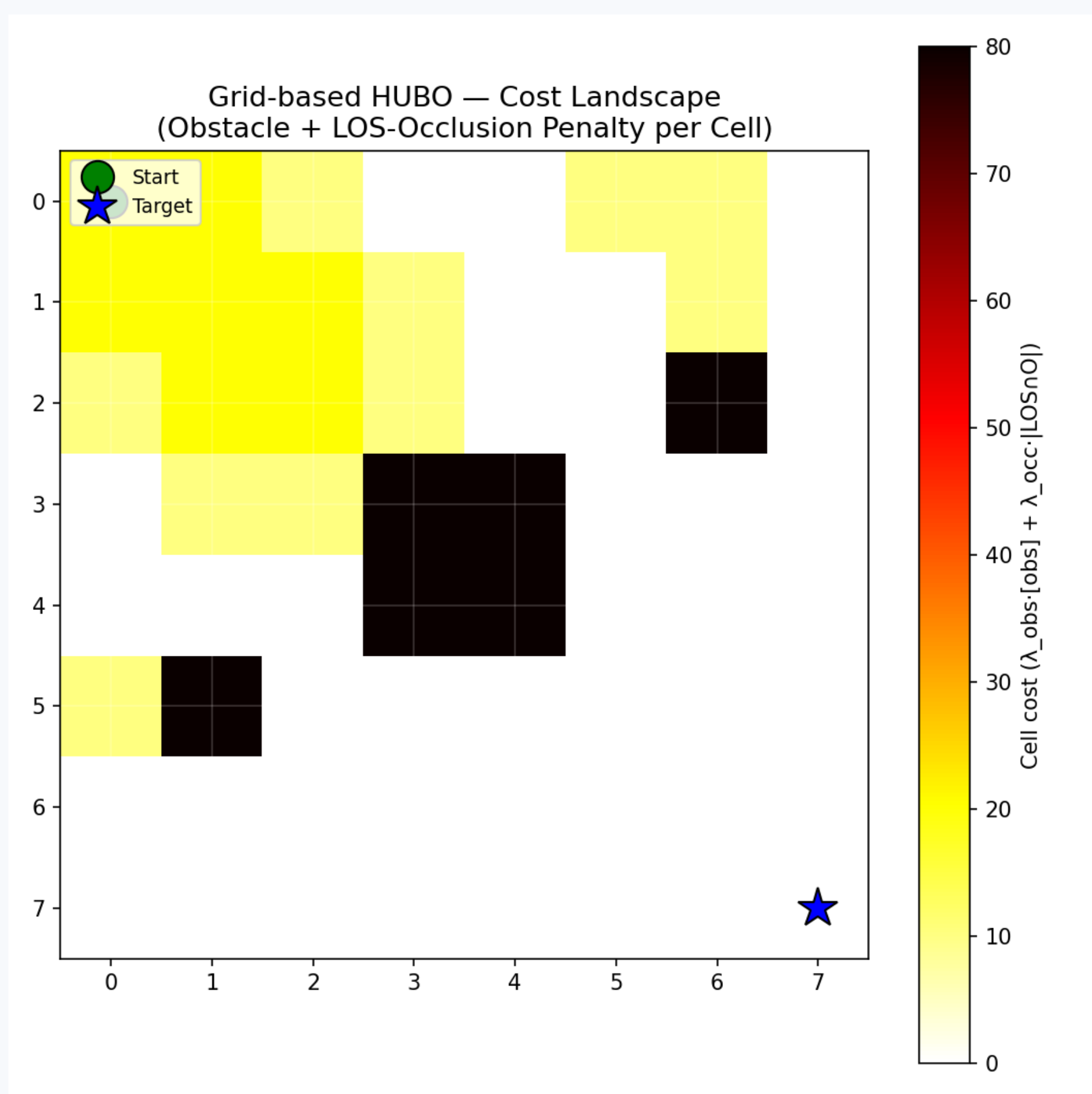
$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_t \sum_u x[t, u] | \text{LOS}(u, \text{tgt}) \cap O |$$

Genuinely cubic proximity (lingering in a buffer cell):

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{v_b} \sum_t x[t-1, v_b] x[t, v_b] x[t+1, v_b]$$

Why HUBO not QUBO? Quadratizing the cubic term needs $\sim 3,172$ ancillas ($\sim 30\%$ more variables) with dense couplings that wreck sparsity.

Occlusion Cost Surface



Yellow = target occluded from that cell; dark = obstacle.

3. Feasibility-Aware Benchmarking

Penalized energy misleads — annealers lower energy by *violating* soft constraints. Four safeguards:

(i) Energy decomposition

hard-penalty vs. soft-cost, kept separate

(ii) Independent constraint check

decode + verify, not energy thresholds

(iii) Matched sampling budgets

equal reads/runs across solvers

(iv) Penalty-weight sweep

exposes the constraint-exploitation regime

Two-Tier Design

State-vector QAOA scales as 2^n — impossible at $n=960$.

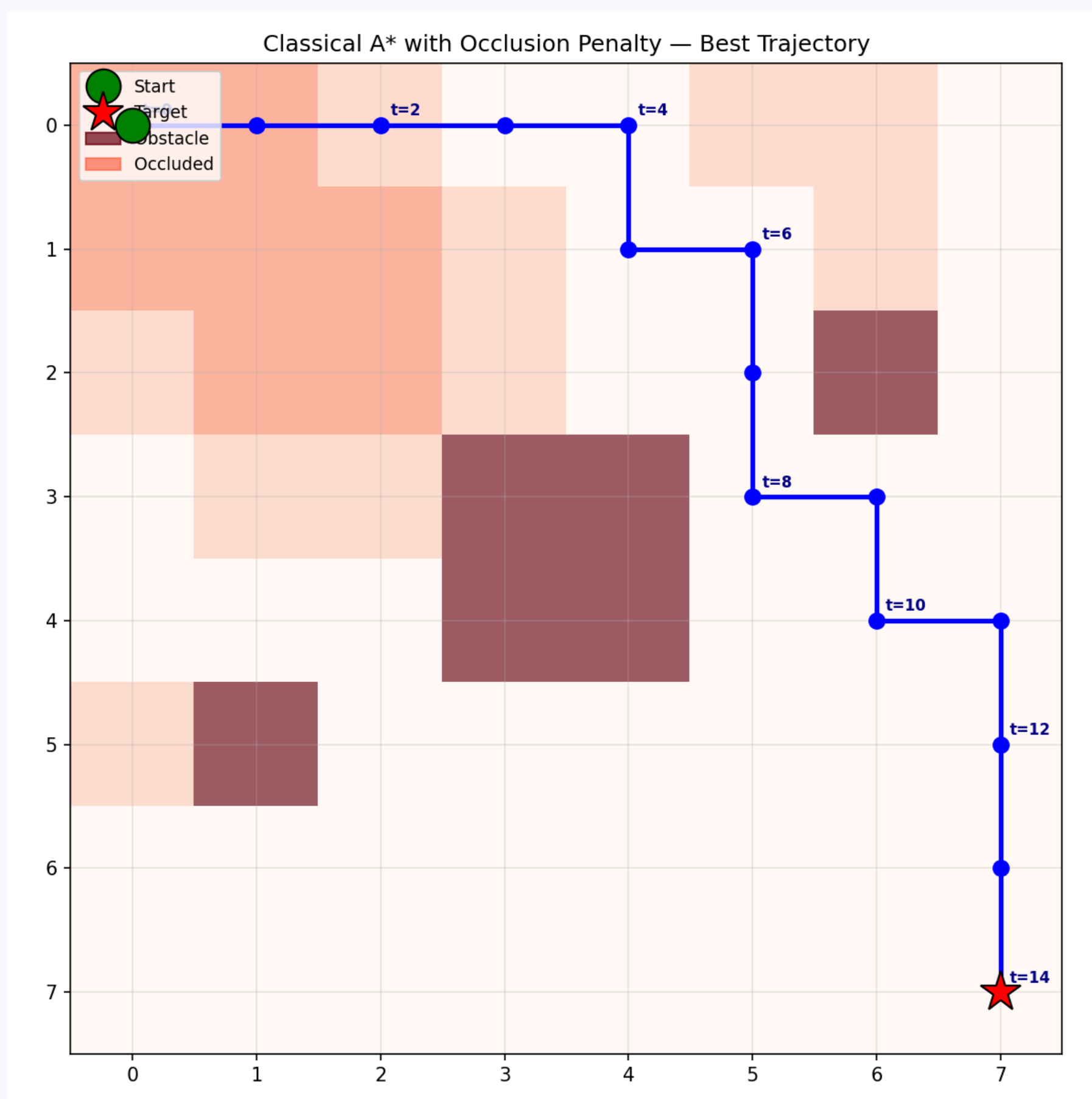
TIER 1	TIER 2
Full 8×8	Tiny 2×2
960 vars	8 vars
$A^* \cdot SA \cdot \text{neal}$	+ Qiskit QAOA

Classical A* Baseline

A* on the same grid with occlusion-weighted cell cost

$$1.0 + \lambda_{\text{occ}} | \text{LOS} \cap O | + \lambda_{\text{prox}} \mathbf{1}[v_b \in B]$$

grounds the HUBO against a conventional planner — answering whether the formulation captures **sensible planning** or is a benchmark artifact.

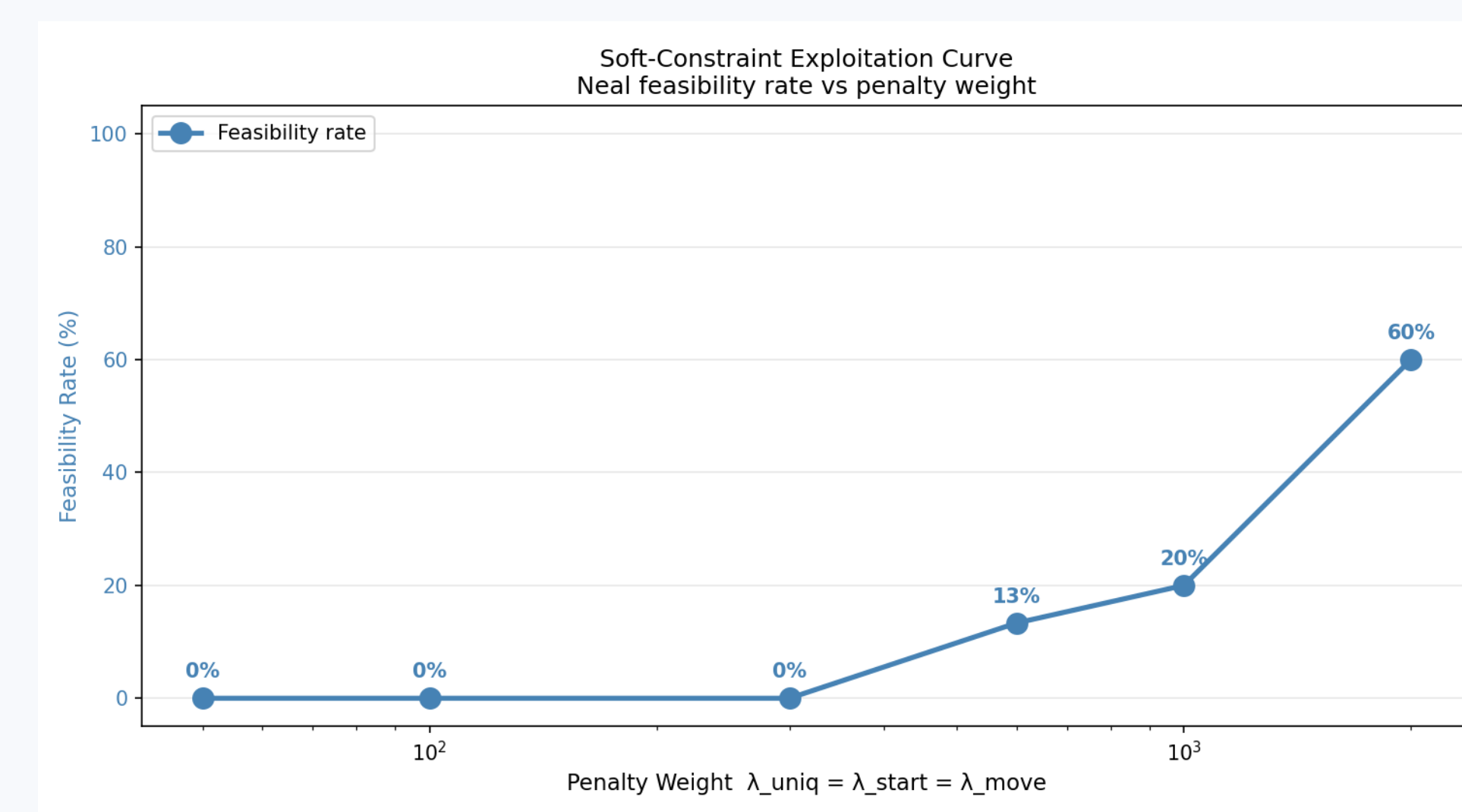


4. Results: 8×8 Instance

Metric	A*	SA	neal
Soft cost	317	303	496
Feasibility	100%	100%	22%
Reaches target	yes	14%	8%
Visibility	80%	—	—
Runtime/sample	10 ms	47.8 s	0.26 s

A* and SA reach comparable quality; A* is $\sim 4,700\times$ faster. neal is fast but only 22% feasible.

Headline: Constraint-Exploitation Curve



neal feasibility climbs **0% → 60%** as λ : 50 → 2000

General-purpose annealers need penalty weights an **order of magnitude** above the objective scale to respect one-hot HUBO structure. A single λ would misreport feasibility.

Tier 2 & Takeaways

On 8 variables, **all four solvers** reach the same optimum ($E^* = -118.17$); QAOA is a correctness check ($2,750\times$ slower, no scaling).

The HUBO's value is not raw speed here, but compatibility with quantum solvers and extensibility to a **moving target**, where occlusion becomes intrinsically cubic — beyond A*'s shortest-path reach.

Future: moving target · real D-Wave hardware · vs. continuous MPC.

CONTACT + DOI REFERENCES

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Selected DOIs: [6] 10.1109/ACCESS.2022.3157977; [7] 10.48550/arXiv.2508.21361; [8] 10.48550/arXiv.2501.18432.